TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME: _____

CIVICS GROUP: _____

H2 MATHEMATICS 9758/01

Paper 1 15 SEPTEMBER 2026

3 hours

Additional materials: Printed Answer Booklet

List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** the questions.

Write your answers in the spaces provided in the Printed Answer Booklet. Follow the instructions on the front cover of the Printed Answer Booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of 7 printed pages and 1 blank page.

1 A curve C has equation $2x^2 + xy + y^2 = 14$.

(a) Show that $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{4x+y}{x+2y}$. [2]

(b) The line L is a normal to the curve C at a point $P(x, y)$. Given that the gradient of L is undefined at P and P lies in the second quadrant, find the coordinates of P. [3]

(c) Line L meets C again at the point Q. Find the coordinates of Q. [3]

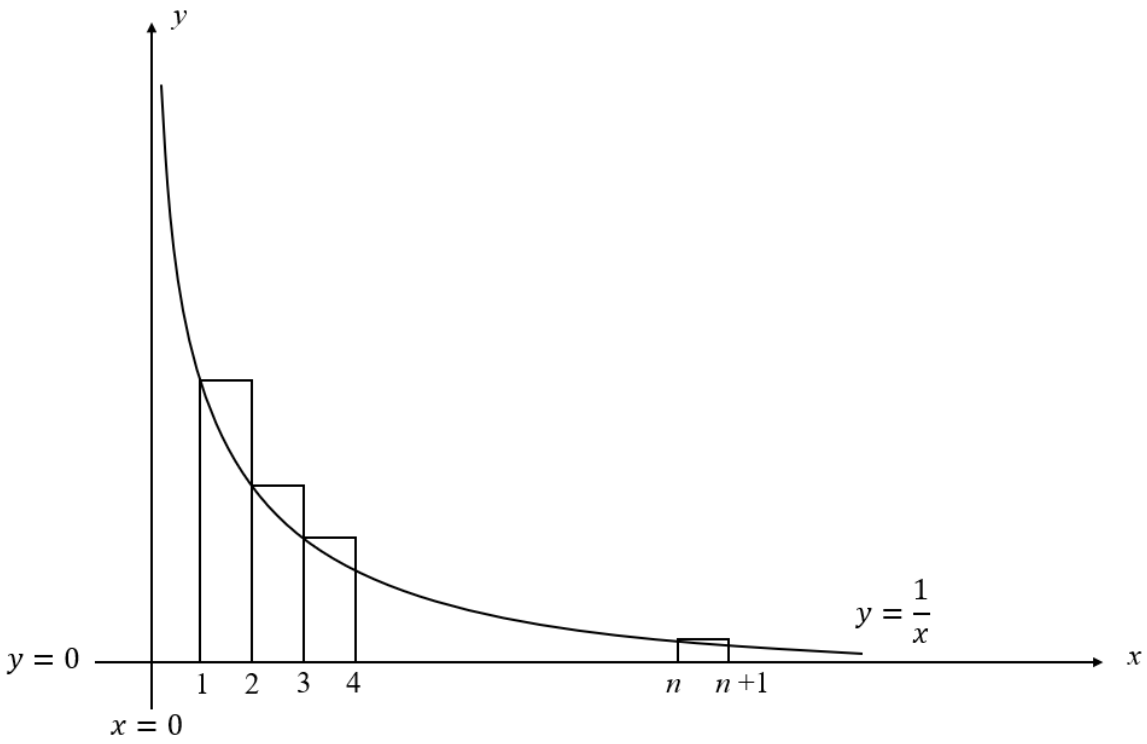
2 **(a)** Using the substitution $u = x^3 + 1$, solve $\int_0^2 \frac{x^5}{\sqrt{x^3+1}}\,\mathrm{d}x$ algebraically. [4]

(b) Find $\int \sin^{-1} x \,\mathrm{d}x$. [3]

3 Show that the differential equation $\left(1+x^2\right)\frac{\mathrm{d}y}{\mathrm{d}x} = y\left(1+y^2\right)$ may be reduced to $\frac{\mathrm{d}\theta}{\mathrm{d}x} = \frac{\tan\theta}{1+x^2}$ by using the substitution $y = \tan\theta$, where $0 < \theta < \frac{\pi}{2}$. Hence, find the particular solution given that $y = 1$ when $x = 0$. [8]

4 **(a)** Given that $\sum_{r=1}^{n} r^2 = \frac{n}{6}(n+1)(2n+1)$, find $\sum_{r=4}^{n}\left[(r-3)(2r-1)\right]$ in terms of n. [4]

(b) The diagram below shows the graph of $y = \frac{1}{x}$, with n rectangles of equal width of 1 unit from $x = 1$ to $x = n+1$.



(i) Write down S_n, the total area of the n rectangles. [1]

(ii) By considering area under the curve using integration, show that the infinite series $S = 1 + \frac{1}{2} + \frac{1}{3} + \ldots$ is divergent. [2]

5 The curve C has equation $y = \dfrac{2x^2 + kx + 13}{x - 2}$. It is given that the line $y = 2x - 4$ is an asymptote of C.

(a) Show that $k = -8$. [1]

(b) Sketch C, stating the coordinates of any points of intersection with the axes, turning points and the equations of any asymptotes. [3]

(c) By considering an appropriate graph and the sketch of C, find an inequality involving a and b such that the equation $\dfrac{(x-2)^2}{a^2} - \dfrac{1}{b^2}\left(\dfrac{2x^2 - 8x + 13}{x - 2}\right)^2 = 1$, where $a > 0$ and $b > 0$, has two distinct real roots. [2]

(d) Given that $a = 1$ and $b = 4$, find the roots of the equation in part **(c)**. [2]

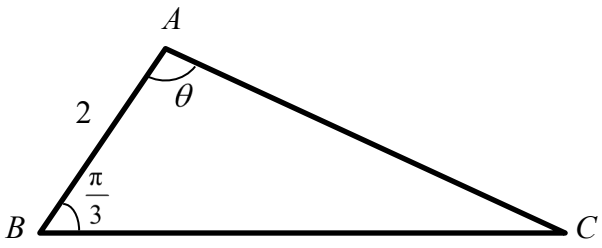
6 The curves C and D have equations $x^2 + \dfrac{y^2}{(\mathrm{e}-1)^2} = 1$ and $y = \mathrm{e}^{x+1} - 1$ respectively.

(a) Sketch C and D on the same diagram, clearly labelling all axial intercepts and the equations of any asymptotes. [4]

The smaller region enclosed by C and D is denoted by R.

(b) Using integration, find the exact volume generated when R is rotated completely through 4 right angles about the x-axis. Express your answer in the form $\left(a\mathrm{e}^2 + b\mathrm{e} + c\right)\pi$ where a, b and c are constants to be determined. [6]

7



In the triangle ABC, $AB = 2$, angle $BAC = \theta$ radians and angle $ABC = \frac{\pi}{3}$ radians (see diagram).

(a) Show that $AC = \dfrac{2\sqrt{3}}{\sqrt{3}\cos\theta + \sin\theta}$. [4]

(b) Given that θ is a sufficiently small angle, express AC in powers of θ up to and including the term in θ^2. [5]

(c) Find the range of θ for the expansion in part **(b)** to be valid. [2]

8 **Do not use a calculator in answering this question.**

The equation $z^2 - (7+\mathrm{i})z + (24+7\mathrm{i}) = 0$, has a root $z_1 = 4 + k\mathrm{i}$, where $k \in \mathbb{R}$.

(a) Show that $k = -3$ and find the other root, z_2. [4]

On the Argand diagram, the points O, Z_1 and Z_2 represent the origin, and the complex numbers z_1 and z_2 respectively.

(b) With reference to O, explain how the point Z_2 can be obtained from the point Z_1 geometrically. [1]

The point Z_3 represents the complex number z_3, where $z_3 = z_2 - z_1$.

(c) Sketch an Argand diagram showing the points Z_1, Z_2 and Z_3, showing clearly the geometrical relationship between the three points. [3]

(d) State the shape of $OZ_1Z_2Z_3$. [1]

(e) It is known that $(z_1)^2$, z_2 and a real number are roots of a polynomial equation of degree n, with all real coefficients. State the smallest possible value of n and explain your answer. [2]

9 Two friends, Aaron and Bob, start their individual long-term savings plans on 1 January 2026. They both make their annual deposits on the first day of each year.

Aaron deposits \$1500 into Account A on 1 January 2026. On the first day of each subsequent year, he increases his deposit amount by a fixed value of \$20 i.e. \$1520 in Year 2, \$1540 in Year 3, etc. This account does not pay any interest, but he receives a fixed bonus payout of \$$b$ credited to his account on 31 December of every year.

Bob deposits \$1500 into Account B on the first day of every year, starting from 1 January 2026. This account pays a compound interest rate of 3% per annum, credited into his account on 31 December of each year.

Let A_n and B_n be the total amount accumulated in Account A and Account B respectively after the interest is added, at the end of n years i.e. on 31 December of the nth year.

(a) Show that

(i) $A_n = 10n^2 + (1490 + b)n$, [3]

(ii) $B_n = 51500\left(1.03^n - 1\right)$. [3]

(b) Find the value of b if the total amount accumulated in Account A at the end of 5 years is \$9200. [2]

For the rest of the question, use the value of b found in part **(b).**

(c) Find the year in which the total amount accumulated in Account B first exceeds the total amount accumulated in Account A. Hence state the difference in their total amount accumulated at the end of that year. [4]

(d) Suppose the compound interest rate for Account B is now revised to x % per annum. Find the minimum value of x, correct to 2 decimal places, such that the total amount accumulated in Account B first exceeds the total accumulated amount in Account A at the end of 8 years. [3]

10 The plane p_1 has equation $x-2y+2z=5$ and the line l_1 has equation $\frac{x-1}{\alpha}=\frac{2-y}{2}=z+3,$ where α is a non-zero real constant.

(a) The acute angle between the plane p_1 and the line l_1 is $\sin^{-1}\frac{4}{9}$. Find the possible values of α. [4]

Use $\alpha=-2$ for the remaining parts of this question.

(b) Find the position vector of the point of intersection, P, between the plane p_1 and the line l_1. [3]

(c) A line l_2 passes through the point $A\left(3,1,4\right)$ and is parallel to the plane p_1. The lines l_1 and l_2 intersect at a point B. Find the coordinates of B. [3]

(d) Find the acute angle between the lines l_1 and l_2. [2]

(e) A variable plane p_2 contains the line l_1 and a point S with coordinates $\left(-3,y,-1\right)$, where $y\in\mathbb{R}$. The plane p_2 is rotated about the line l_1. Given that every possible orientation of the plane p_2 contains the point S, determine the value of y. [3]

End of Paper